TECHNICAL REPORT

Multilingual E-Sahulat AI

Evidence-grounded passport answers in English, Urdu and Roman Urdu

ای سہولت

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Author	Saifullah Waseem
Context	University project, Multilingual E-Sahulat AI hackathon challenge
Domain	Pakistani ordinary passport services (DGI&P)
Pipeline version	`d089d88a44a2` (code + configuration + knowledge-base fingerprint)
Dense model	`intfloat/multilingual-e5-base`
Local LLM	Qwen2.5 3B (`qwen2.5:3b`, via Ollama), optional
Date	October 25, 2026

In one paragraph

E-Sahulat answers passport questions only with word-for-word quotes from 30 official DGI&P passages. A query in any of three scripts is normalized, mapped to shared concepts and matched against the passages. A cascade of rules and tuned thresholds then decides whether to *answer*, *ask for clarification* or *abstain*. Every answer is checked by a citation verifier and shown in a web app as an "attested copy" with a decision stamp. An optional local Qwen model can add a one-line summary written only from the quotes. On a held-out blind set of 47 questions, the final system reaches 72.3% decision accuracy and cuts false answers on unanswerable questions from 57.1% to 42.9%. Everything runs on one laptop with no external API.

At a glance

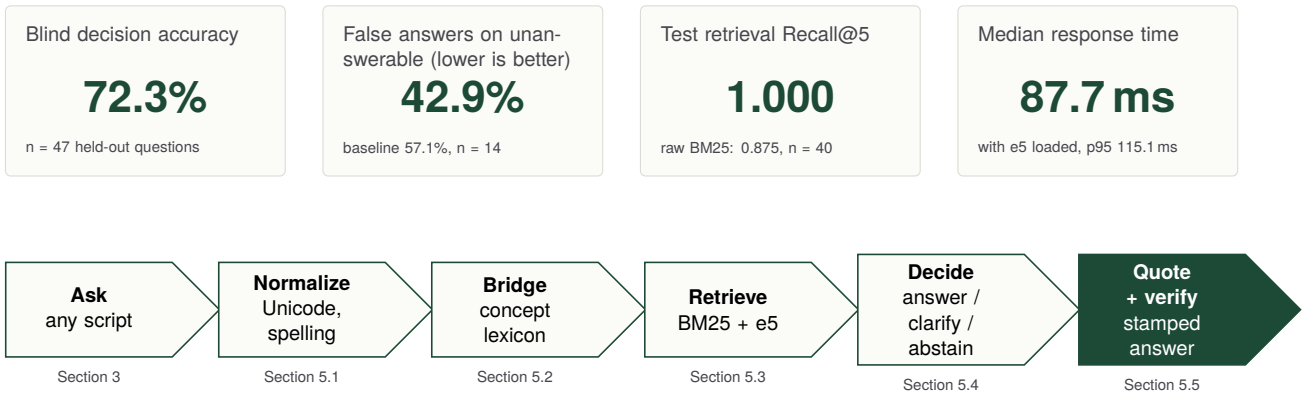


Figure 1. The six stages every question passes through, and where each is explained in this report.

What makes this system different from a chatbot. It never writes its own answer. It finds the official sentence, proves the quote is exact, and says so with a stamp. When no sentence supports an answer, it says "not on record" instead of guessing. The table below shows the three possible outcomes; each has its own stamp in the interface (Section 7).

Stamp	Decision	Meaning for the citizen
VERIFIED TRUE COPY	`answer`	An exact quote from a DGI&P page answers the question; the passage code and source link are shown.
PARTICULARS REQUIRED	`clarify`	The question could mean several services (e.g. "how much does it cost?"); the user is asked which one.
NOT ON RECORD	`abstain`	The official passages don't support an answer, the question is out of scope, or the request is unsafe.

Table 1. The three decisions and their stamps.

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1 Introduction

1.1 The problem

Pakistan's passport rules live on official web pages written mostly in English. Citizens ask about them in Urdu script, in Roman Urdu (Urdu written in Latin letters, with no fixed spelling), in English, or in a mix of all three. The same question can arrive in very different forms:

Style	The same question
English	Which documents are required to renew a passport?
Urdu script	پاسپورٹ کی تجدید کے لیے کون سی دستاویزات درکار ہیں؟
Roman Urdu	Passport renew karwanay ke liye kya documents chahiye?
Noisy Roman Urdu	pasport renu karwana hy
Code-mixed	Passport renewal ke liye online apply karna hota hai ya office jana hota hai?

Table 2. One intent, five surface forms. Keyword search fails on most of them because the official text is English.

Two failure modes make this hard. **Retrieval** breaks when the query's script or spelling differs from the source. **Generation** is dangerous: a fluent chatbot can confidently invent a fee, a document or a processing time, and a citizen may act on it.

1.2 Goals

The challenge brief asks for an assistant that (1) retrieves the correct official evidence regardless of script, spelling or paraphrase, (2) answers only what that evidence supports, with a traceable citation, and (3) abstains or asks for clarification when evidence is missing, all measured on a documented test set with no external API calls.

1.3 Contributions

- A **cross-script concept lexicon** that maps English, Urdu and Roman Urdu surface forms to shared concepts, so an Urdu query can match English evidence (Section 5.2).
- A comparison of **eleven retrieval configurations**: keyword, character n-gram, multilingual dense (e5) and hybrid, with the production choice made on development data only (Section 5.3).
- A **decision cascade** of rule gates plus dev-tuned thresholds that returns *answer*, *clarify* or *abstain* (Section 5.4).
- **Extractive answering with a citation verifier**: every displayed claim is a verbatim official quote whose ID, URL and numbers are checked (Section 5.5).
- An optional **local LLM layer** (Qwen2.5 3B) that writes a short summary from the verified quotes only, with an automatic check for invented numbers (Section 5.6).
- A **web application** designed around the "attested copy" metaphor, bilingual and right-to-left aware (Section 7).
- An **honest evaluation**: a blind set frozen before the final fixes, bootstrap confidence intervals and a leakage audit (Sections 8–10).

1.4 How the brief's requirements are met

Requirement	How E-Sahulat meets it	Section
FR-1 Multilingual input	Urdu script, Roman Urdu, English and code-mixed queries; script detected automatically	5.1
FR-2 Normalization	NFKC, Arabic→Urdu letters, digits, elongation, fuzzy spelling repair	5.1
FR-3 Two+ retrievers compared	11 configurations: BM25, char TF-IDF, e5 dense, hybrids, rerankers	5.3, 9
FR-4 Evidence-grounded answers	Verbatim quote with passage code, title and DGI&P link	5.5
FR-5 Unanswerable queries	"Not on record" abstention and "particulars required" clarification	5.4
FR-6 Traceability	Passage IDs, document IDs, URLs and text hashes kept end to end; decision trace in the UI	2, 7
FR-7 Evaluation	Recall@5, MRR@5, citation support, abstention, cross-script, latency, with n and CIs	8, 9
FR-8 Usability	Web app with language choice, evidence view, knowledge base and evaluation pages	7

Table 3. Functional requirements from the challenge brief.

2 Data and provenance

2.1 The organisers' dataset

The hackathon supplied one file, `Multilingual_E-Sahulat_AI_Hackathon_Dataset.docx`, holding 150 JSON records: 90 `train_dev` records (30 intents × English, Urdu and Roman Urdu) and 60 `test` records (30 hard cases `N001`–`N030` and 30 noisy Roman Urdu spelling variants `SV001`–`SV030`). Each answerable record carries the official evidence excerpt in English and Urdu and its `dgip.gov.pk` source URL.

2.2 The knowledge base

The 30 distinct evidence excerpts (`F01`–`F30`) form the corpus. They come from 5 DGI&P pages. Each passage keeps:

Example passage record (F12)

```
pid: F12   doc_id: D2   topic: minor_validity   authority: official
url: https://dgip.gov.pk/passport/ordinary-passport.php
text: Children under 15 years of age will be issued Passport having 05 years validity
only.
text_ur: 15 سال سے کم عمر بچوں کو صرف پانچ سال کی میعاد والا پاسپورٹ جاری ہوگا۔
fact_key: validity_children_under_15 = 5 years   sha256: (16-hex prefix)   freshness:
needs_manual_review
```

The **English excerpt is always the cited text**. The Urdu translation is indexed (so Urdu queries can match it) and shown beside the quote, but it is the organisers' translation, not an official one.

2.3 Query sets

The official `train_dev` split has no unanswerable questions, which makes it impossible to tune an abstention threshold. 49 team-written development queries were therefore added (out-of-domain, other services, ambiguous, personal, noise, prompt injection, unsafe, and confusable siblings),

used *only* for tuning. A separate blind set of 47 questions was written and its hash frozen before the final round of fixes; it is the primary result.

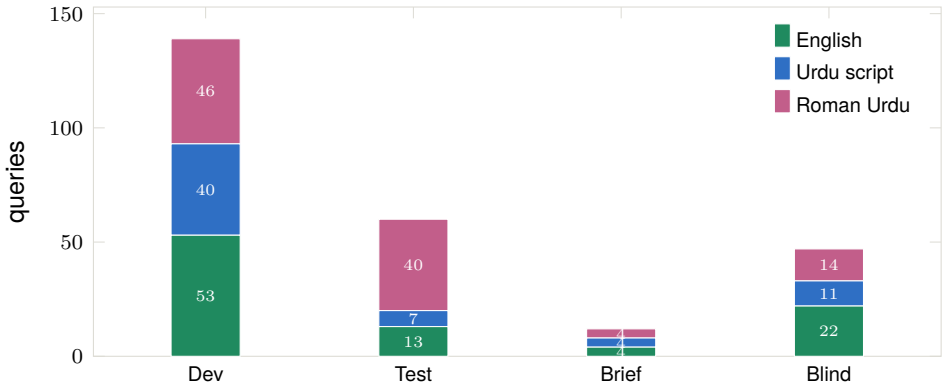


Figure 2. Queries per split and language style (258 in total). Dev = 90 official + 49 team-written; Test = official `test` (60); Brief = the challenge's 12 validation cases; Blind = held-out set used once.

Split	Source	n	Use
Dev	official `train_dev` + team queries	139	Lexicon design, retriever selection, threshold tuning
Test	official `test`	60	Seen during development, so used as a regression check only
Brief	the brief's 12 validation cases	12	Team-labelled; reported separately
Blind	written by the team, hashes frozen	47	Run once on the final pipeline; the honest measure

Table 4. How each query set is used. The test split was never used for training or threshold tuning.

3 System architecture

The system has three layers: a browser front end, a Python server that hosts the pipeline, and an optional local LLM. Figure 3 shows how they talk to each other; Figure 4 zooms into the pipeline.

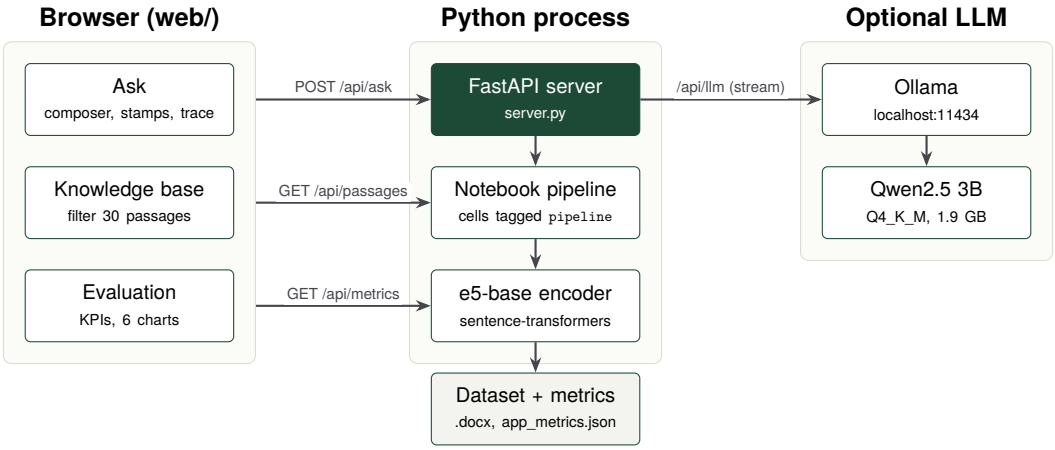


Figure 3. Deployment. The server loads the notebook's pipeline cells at start-up, so the app runs exactly the evaluated code. Nothing leaves the machine: Ollama and the e5 model both run locally.

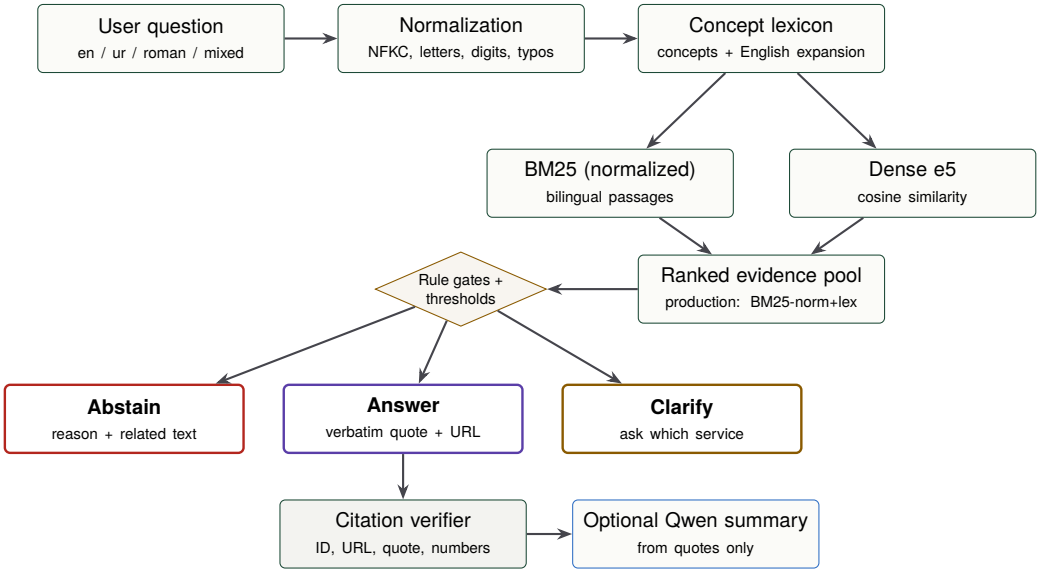


Figure 4. The answering pipeline. The dense score feeds the decision thresholds even though the production *ranking* is keyword-based (Section 5.3).

4 A question's journey, step by step

To make the pipeline concrete, here is what actually happened to one Roman Urdu question in the running system (pipeline `d089d88a44a2`). Every value below was read from the system's decision trace.

1	**Input**	"Passport renew karwanay ke liye kya documents chahiye?"
2	**Script detection**	Roman Urdu (Latin letters with Urdu function words such as *ke liye*, *kya*).
3	**Normalization**	Lower-cased, punctuation and spacing cleaned. No spelling repair was needed for this query.
4	**Concepts found**	`DOCUMENT`, `PASSPORT`, `RENEW`, `REQUIRE`. *renew karwanay* and *chahiye* are Roman Urdu forms the lexicon maps to English concepts.
5	**Retrieval (top 5)**	F17, F16, F05, F18, F09. F17 is "Online Passport Renewal Service is available for both Inland & Overseas Pakistanis."
6	**Evidence scores**	Dense similarity $0.8475 \geq \tau_d = 0.8231$; keyword coverage $0.40 \geq \tau_c = 0.0$.
7	**Rule gates**	No injection, unsafe, personal, out-of-scope or ambiguity cue fires.
8	**Decision**	VERIFIED TRUE COPY `answer`, reason `supported`.
9	**Verifier**	Quote is a substring of F17, URL matches the document registry, no numbers to check: pass.
10	**Response time**	about 1 ms when the query embedding is cached; about 70–90 ms for a new query with e5.

Figure 5. A real trace through the system. Steps 4 and 5 are where the cross-script bridge does its work: the query contains no English word from F17 except "passport", yet F17 ranks first.

The same intent in Urdu script

The Urdu-script version, پاسپورٹ کی تجدید کے لیے کون سی دستاویزات درکار ہیں؟, produced the same four concepts, the same top-5 list (F17, F16, F05, F18, F09) and the same answer (F17), with dense similarity 0.8466. This is the cross-script consistency the lexicon is designed for.

5 Methods

5.1 Query normalization

Queries and passages go through the same meaning-preserving normalization, so both sides of the match look alike.

Step	What it does	Example
Unicode	NFKC, invisible characters removed	full-width or decomposed letters become standard
Letters	Arabic-keyboard *yeh*, *kaf*, *heh* unified to Urdu code points	ي → ی, ك → ک
Marks	diacritics and *tatweel* removed; Urdu punctuation mapped	پاسپـورٹ → پاسپورٹ
Digits	Urdu and Arabic-Indic digits to ASCII	۱۵ → 15
Roman Urdu	lower-case, letters repeated 3+ times collapsed, digit-letter split	*10saal* → *10 saal*, *pleeease* → *please*
Numbers	number words to digits	*das*, *ten* → 10
Spelling	fuzzy repair of unknown Latin tokens (below)	*documnts* → *documents*

Table 5. Normalization steps.

Spelling-variant repair. A Latin token w of at least five characters that the lexicon does not know is replaced by the closest known form, but only if it is close enough:

$$w^* = \arg\max_{v \in V_{\text{lex}}} \text{sim}(w, v) \qquad \text{if} \quad \max_{v} \text{sim}(w, v) \geq 0.84, \tag{1}$$

where sim is the Ratcliff–Obershelp similarity ratio (Python's `difflib`). Known tokens are never rewritten, and multi-word forms such as *fst track* are matched as phrases.

5.2 Cross-script concept lexicon

More than fifty concepts list their English, Roman Urdu and Urdu-script surface forms. A query q yields a concept set $C(q)$ and an expansion $E(q)$ of English terms taken from the corpus vocabulary; keyword retrievers then score $q \oplus E(q)$.

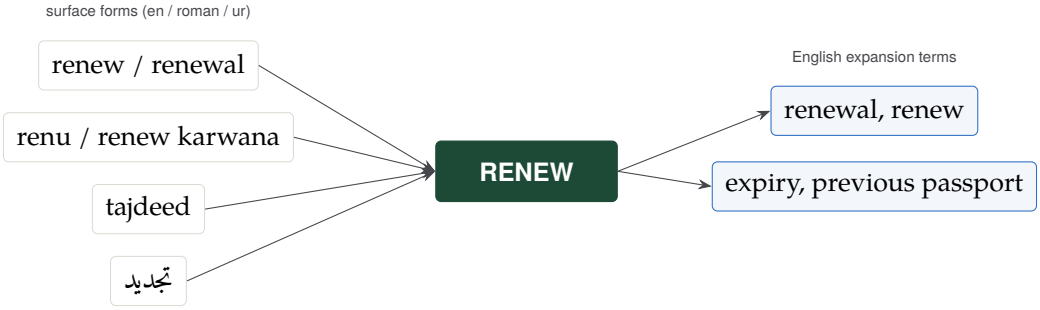


Figure 6. How one concept bridges scripts. Many surface forms map to `RENEW`, which adds English terms the official passages actually use. Illustrative forms; the full lexicon is in the notebook (cell 8).

The lexicon also encodes **number phrases**, because several passages differ only by a number: "*N saal ka passport*" means validity (F11/F12), while "*N saal ka bacha*" means the applicant's age (F06/F07).

5.3 Retrieval

Keyword retrieval (BM25). Passages are scored with BM25Okapi ($k_1 = 1.5$, $b = 0.75$):

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, d)\,(k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b\,\frac{|d|}{\overline{|d|}}\right)}, \tag{2}$$

where $f(t, d)$ is how often token t occurs in passage d and $\overline{|d|}$ is the mean passage length. `BM25-raw` uses raw text; `BM25-norm+lex` uses normalized text plus the lexicon expansion; the `+bi` variants index the Urdu translation alongside the English passage.

Character n-grams. TF-IDF over character 3–5-grams (scikit-learn), a spelling-robust baseline.

Dense retrieval. The query and each passage are embedded with `intfloat/multilingual-e5-base` using the model's `query:` and `passage:` prefixes and compared by cosine similarity:

$$s_{\text{dense}}(q, d) = \frac{\mathbf{e}_q \cdot \mathbf{e}_d}{\|\mathbf{e}_q\| \, \|\mathbf{e}_d\|}. \tag{3}$$

Hybrid fusion. Rankings are fused with Reciprocal Rank Fusion, weighted by source authority $w_{a(d)}$ (1.0 official, 0.8 third-party):

$$s_{\text{RRF}}(d) = w_{a(d)} \sum_{r \in R} \frac{1}{60 + \text{rank}_r(d)}. \tag{4}$$

Concept re-ranking. The `+rerank` variants add $\lambda \cdot \text{agree}(C(q), C(d))$ to separate sibling passages that share most words (e.g. Normal / Urgent / Fast Track processing in F13–F15); λ is tuned on dev.

Selection. All eleven configurations are run on dev, and the one with the best dev MRR@5 becomes the production retriever. With e5 this was again **BM25-norm+lex**, which reached a perfect dev MRR@5 (Figure 16).

5.4 Decision cascade

Each query gets exactly one decision through ordered gates. Rules handle the cases a similarity score can't (a prompt injection can be highly similar to a real question); thresholds handle the rest.

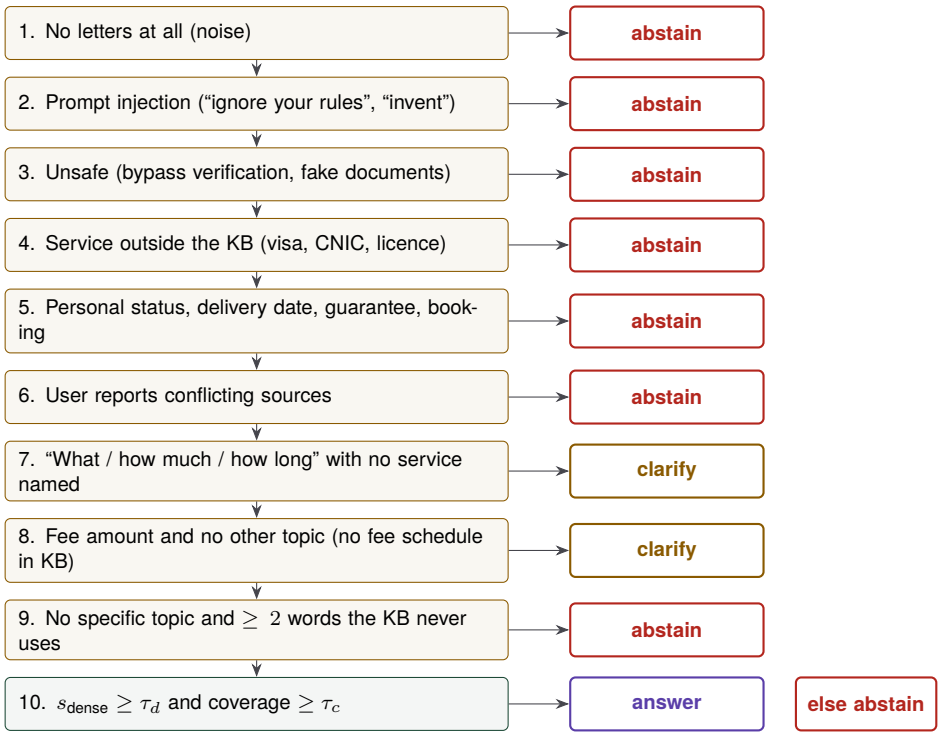


Figure 7. The decision cascade. The first gate that fires decides; gate 10 is reached only by clean, in-domain questions.

Threshold tuning. The thresholds are grid-searched on dev only, maximizing an objective that penalizes false answers more than missed answers:

$$J(\tau) = 0.4 \cdot \text{Recall}_{\text{answer}}(\tau) + 0.6 \cdot \text{Acc}_{\text{non-answer}}(\tau), \tag{5}$$

taking the middle of the best plateau for robustness. With e5 this gives $\tau_d = 0.8231$ and $\tau_c = 0.0$. Figure 8 shows why a single cut can't be perfect: the two groups overlap.

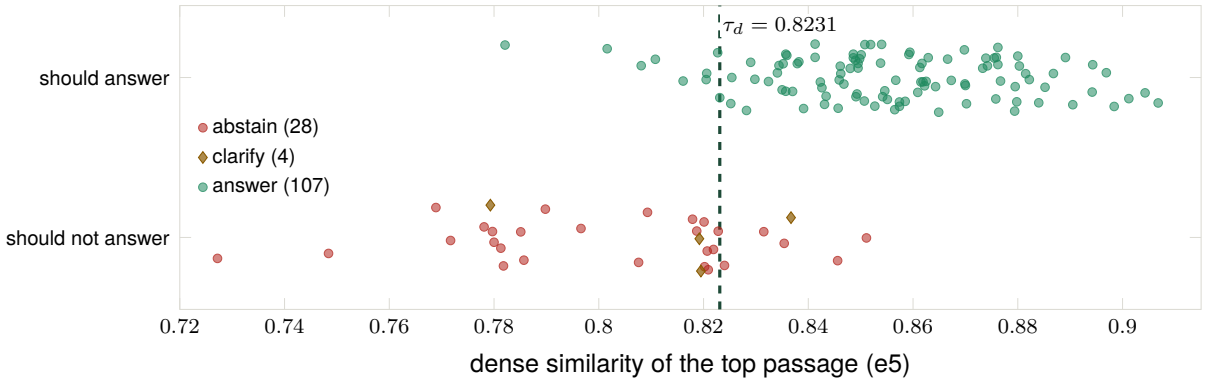


Figure 8. Dev questions by their top dense score, jittered vertically. Left of the dashed line the system abstains. 8 answerable questions fall below the threshold (missed answers) and 6 non-answerable ones sit above it; those are caught by the rule gates or by low coverage instead.

5.5 Grounded answers and citation verification

Answers are **extractive**: the top official passage is quoted word for word with its passage code, page title and URL. Before anything is shown, a verifier checks every evidence item:

(a) the passage ID exists in the knowledge base;

(b) the URL equals the document registry entry for that passage;

(c) the quoted span is an exact substring of the passage;

(d) every number in the span appears in the passage.

Tamper tests in the notebook (altered quote, wrong URL, unknown ID) confirm that the verifier rejects fabricated citations. In the interface a failed check replaces the violet stamp with a red "copy not verified" stamp.

Conflict handling. Passages that state the same quantity share a `fact_key`. If the evidence pool holds two different values for one key, the response prefers the official (and, where dated, newer) source and discloses the disagreement. The official KB has no contradictions, so this is demonstrated with a clearly labelled synthetic passage that is never used in any metric.

5.6 Optional local LLM summary

Verbatim quotes are safe but can read stiffly. When the *Qwen summary* switch is on, a local Qwen2.5 3B model rewrites the already-verified quotes as one to three plain sentences. It is boxed in on every side:

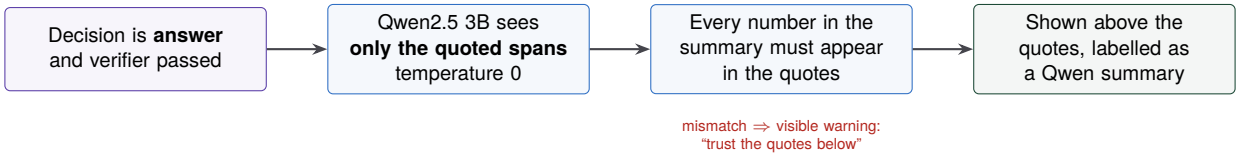


Figure 9. Guardrails around the LLM. It runs only after the pipeline has already decided to answer, so it can never change a decision, add a source, or answer a question the evidence doesn't support.

The system prompt instructs the model to keep every number, fee, document name and time period exactly as written, to mention nothing not in the quotes, and to end each sentence with its passage ID. In testing, the summary for "Can a child under 15 get a ten-year passport?" was: *"No, children under 15 years of age will be issued Passport having 05 years validity only [F12]."* The first summary after start-up took about 26 s (model load); later ones took 8–17 s on CPU.

6 Models used

	Dense encoder	Local LLM (optional)
Model	`intfloat/multilingual-e5-base`	Qwen2.5 3B Instruct (`qwen2.5:3b`)
Role	Embeds queries and passages; its similarity drives the answer/abstain threshold and the dense and hybrid retrievers	Rewrites verified quotes into a short plain-language summary
Architecture	XLM-RoBERTa encoder: 12 layers, 12 heads, 768-dim, 250k-token vocabulary	Qwen2 decoder, 3.1B parameters, 2048-dim embeddings, 32k context
Size on disk	about 1.1 GB (~278M parameters)	1.9 GB (Q4_K_M quantization)
Runtime	sentence-transformers 6.2 on PyTorch 2.14 (CPU)	Ollama, `localhost:11434`, temperature 0, at most 200 tokens
Languages	about 100, including Urdu	multilingual, including Urdu and English
Licence	MIT	Qwen Research License
Why chosen	Strong multilingual retrieval at a size that runs on a laptop CPU; no fine-tuning needed	Smallest capable instruction model that fits next to e5 in laptop memory

Table 6. The two neural models. No model was trained from scratch; "training" here means tuning the retriever choice, re-ranking weight and decision thresholds on the dev split.

A note on the stand-in encoder

An earlier round of the project could not download e5, so its numbers came from a character n-gram LSA stand-in (`841339e5e722`). For this report the notebook was re-run end to end with the real e5 model (`d089d88a44a2`); every e5 number here comes from that run, and Section 11 compares the two.

6.1 Software stack

Layer	Technology
Pipeline	Python 3.14, NumPy, pandas, scikit-learn, own BM25, `difflib`, `unicodedata`, python-docx
Neural	sentence-transformers + PyTorch (CPU) for e5; Ollama for Qwen
Server	FastAPI + Uvicorn (`server.py`): `/api/status`, `/api/ask`, `/api/passages`, `/api/metrics`, `/api/llm`, `/api/llm/start`
Front end	Hand-written HTML, CSS and JavaScript (no framework); SVG charts drawn from the metrics file
Evaluation	Bootstrap confidence intervals (2000 resamples), Matplotlib figures, generated LaTeX tables
Report	LaTeX (XeTeX via Tectonic), TikZ and pgfplots, figures generated from the run's outputs

Table 7. Software used.

7 The interface and its visual design

7.1 Design idea: the attested copy

In Pakistan, an official document is trusted when it carries an attestation: a photocopy stamped and signed as a true copy. The interface borrows that ritual. There is no chat thread. Each question produces one *document*: the question as its heading, a rubber stamp with the decision, and the official sentence laid on it as a paper strip cut from the source.

Element	Visual treatment	Why
Shell	bottle-green "file cover" top bar (■ #1D4A37)	reads as an official file, not a consumer app
Ground	cool photocopy white (#F3F3F0)	paper without the warm "cream" cliché
Decisions	three stamp inks: violet, ochre, red	each outcome is recognizable at a glance; inks are reserved for decisions only
Quotes	serif text on a white strip with its own shadow	the official words look like an excerpt laid on file
Urdu	Nastaliq script, right to left, beside the English	every script is first-class, not a translation footnote
Passage codes	boxed green codes (F12) that link to the knowledge base	traceability with one click
Charts	green, blue and rose series, validated for colour-blind readers	chart colours never borrow a stamp ink

Table 8. Visual vocabulary of the interface.



Figure 10. The palette. Stamp inks (violet, ochre, red) are used only for decisions; the chart series were checked with a colour-vision-deficiency validator.

7.2 Screens

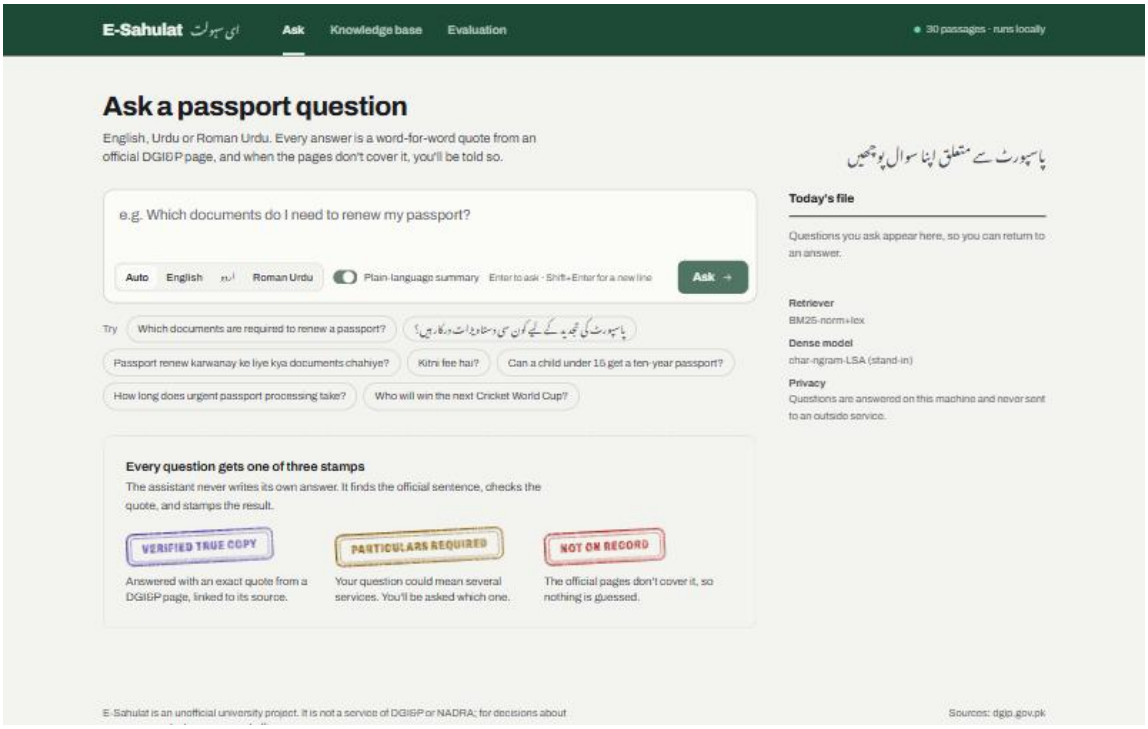


Figure 11. First screen. The composer accepts any script (it switches to right-to-left when Urdu is typed). Below it, the three stamps explain the possible outcomes before the first question; the right column keeps "Today's file" of asked questions.

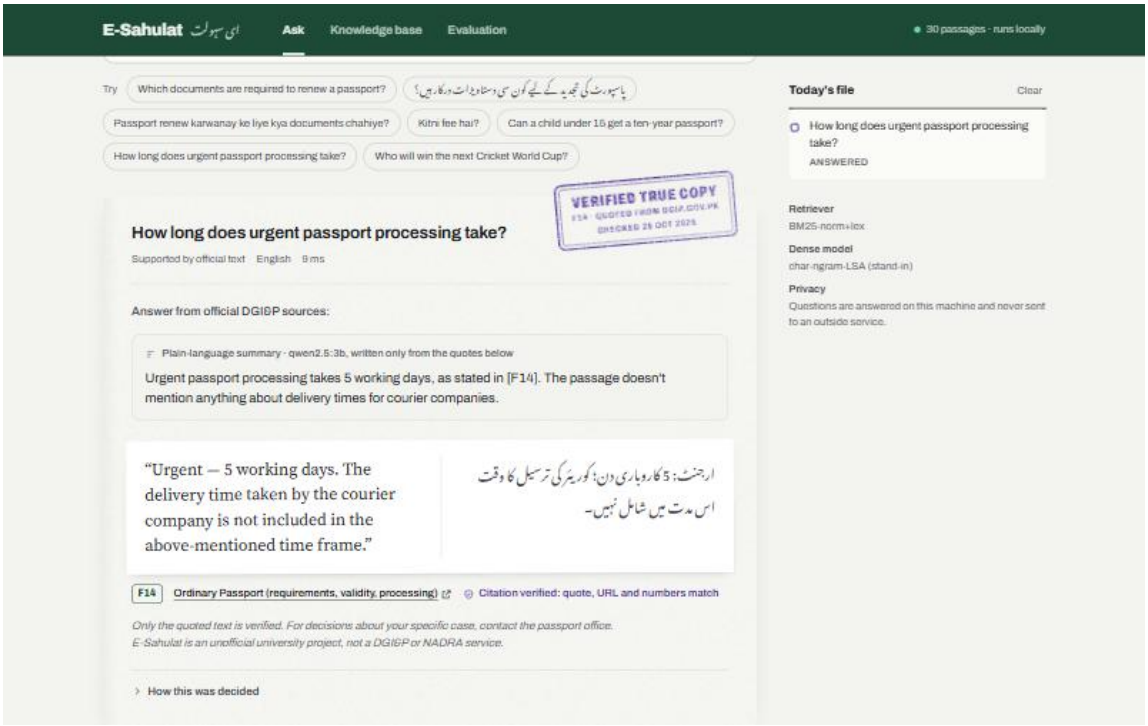
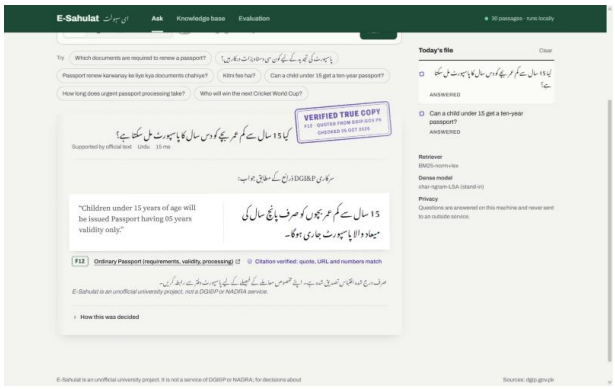
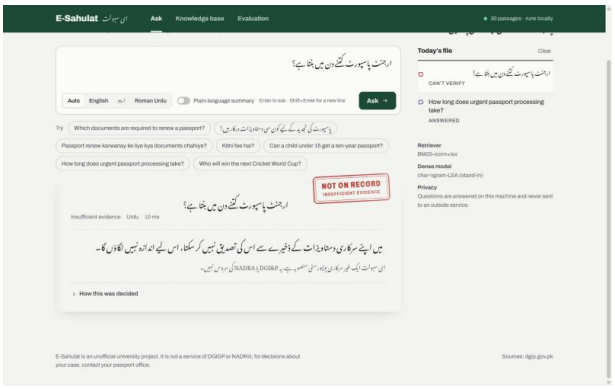


Figure 12. An answered question. The violet stamp carries the passage code and check date. The Qwen summary sits above the verbatim quote; the quote's Urdu text is mirrored across a centre rule; the source link and "citation verified" line follow.

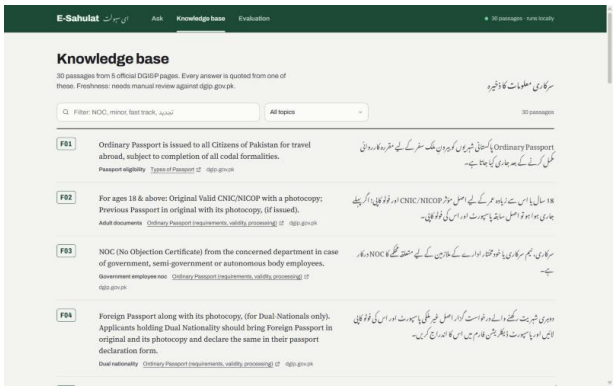


(a) Urdu question: the Urdu text leads, on the right.

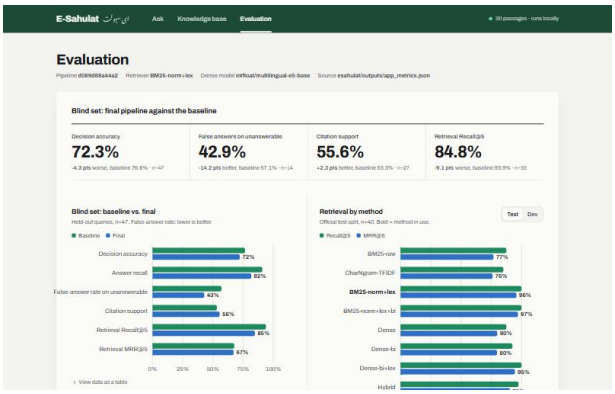


(b) An unsupported Urdu question gets "Not on record".

Figure 13. Urdu is first-class: the layout mirrors for Urdu queries, and messages and disclaimers switch to Urdu.

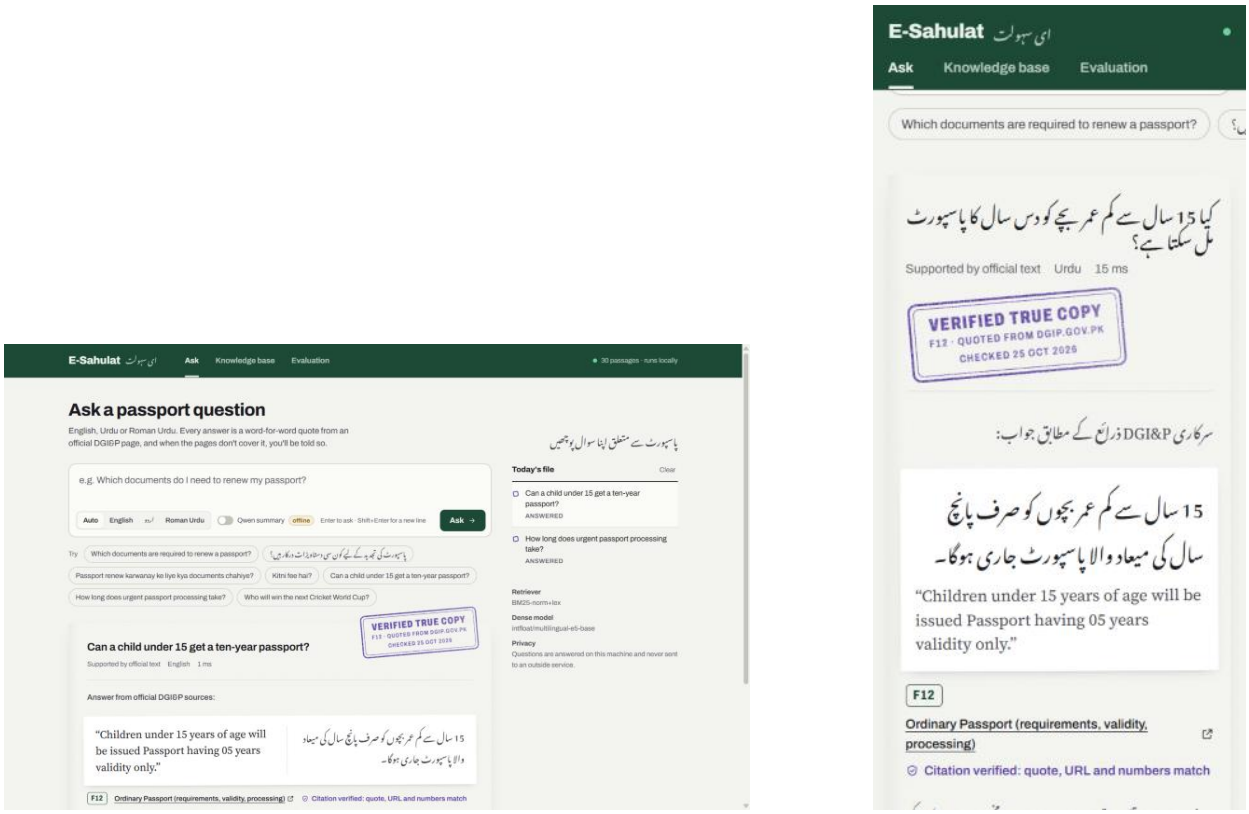


(a) Knowledge base: search and topic filter over the 30 passages.



(b) Evaluation: blind-set KPIs against the baseline.

Figure 14. The two supporting sections. Every chart on the Evaluation page has a "view data as a table" alternative.



(a) Composer with the Qwen switch, here in its "offline" state. **(b)** Phone width (390 px).

Figure 15. Controls and the phone layout. The Qwen switch shows the model's state (ready, starting, offline, model missing) and starts Ollama when switched on.

7.3 Interaction and accessibility

- **One signature motion.** The stamp lands on the document in 200 ms with a strong ease-out curve; everything else uses short (≤ 260 ms) transitions on transform and opacity only, and all motion is reduced under the operating system's reduced-motion setting.
- **Honest states.** A loading banner while the pipeline boots, an empty state that teaches the three outcomes, a clear error when a request fails, and a visible state tag on the Qwen switch.
- **Accessibility.** WCAG AA contrast, full keyboard use, focus rings, `lang` and `dir` attributes on Urdu text, a table view behind every chart, and colour never used alone to carry meaning.
- **Responsible framing.** Every answer and the footer state that this is an unofficial university project, not a DGI&P or NADRA service.

8 Evaluation protocol

For answerable queries Q_{ans} with gold passages G_q and ranking π_q:

$$\text{Recall@5} = \frac{1}{|Q_{\text{ans}}|} \sum_q \mathbb{1}[G_q \cap \pi_q[1..5] \neq \emptyset], \quad \text{MRR@5} = \frac{1}{|Q_{\text{ans}}|} \sum_q \frac{1}{\text{rank}_q} \quad \left(\tfrac{1}{\text{rank}_q} = 0 \text{ if } \text{rank}_q > 5\right). \tag{6}$$

- **Decision accuracy**: share of queries whose decision (answer / clarify / abstain) matches the label.
- **False-answer rate**: share of unanswerable queries that were answered anyway. Lower is better; this is the metric that protects citizens.

- **Citation support**: share of answered queries whose citation passes the verifier *and* points to a gold passage.
- **Cross-script consistency**: whether the English, Urdu and Roman Urdu versions of one intent retrieve the same top passage.
- **Latency**: end-to-end `respond()` time with the model loaded and per-query caches cleared.

Confidence intervals are 95% percentile bootstraps over queries (2000 resamples). Because the official test queries were visible during development, the **blind set is the primary result** and the official test is a regression check.

9 Results

9.1 Retrieval

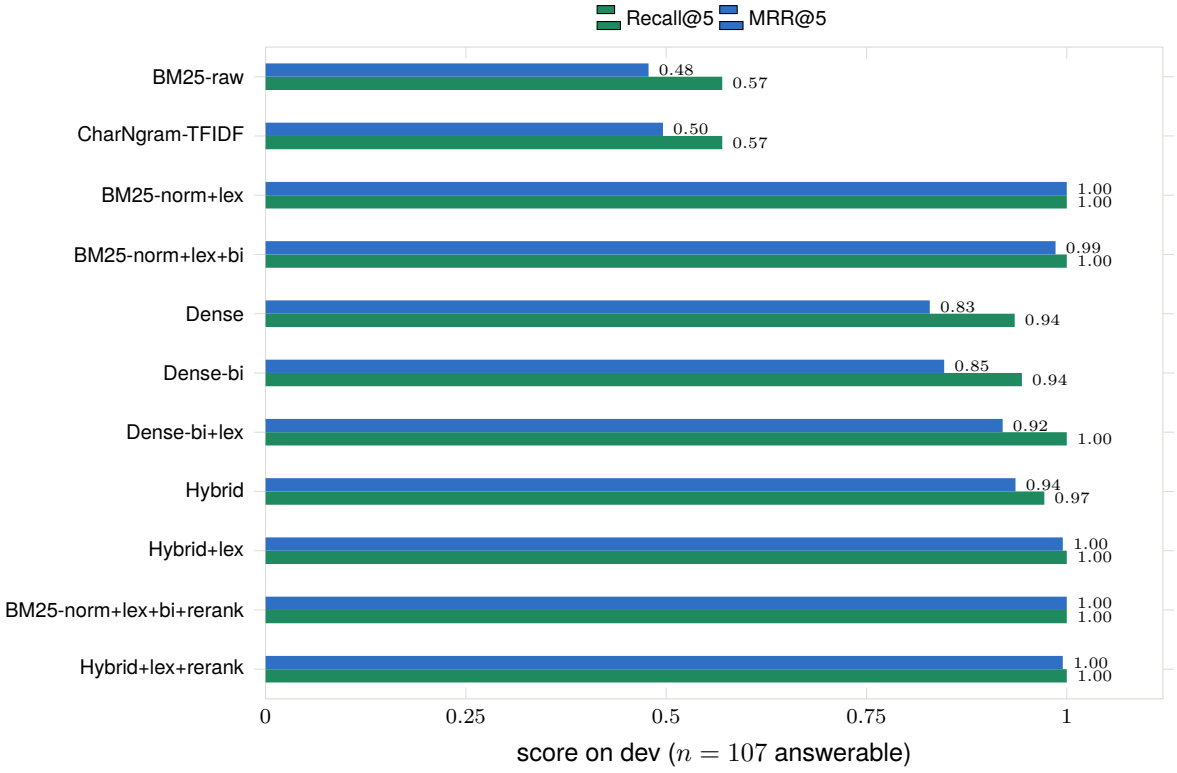


Figure 16. Retrieval on the development split, where the production retriever is chosen. Normalization plus the lexicon lifts BM25 from 0.48 to 1.00 MRR@5; BM25-norm+lex is the first configuration to reach the top, so it was selected.

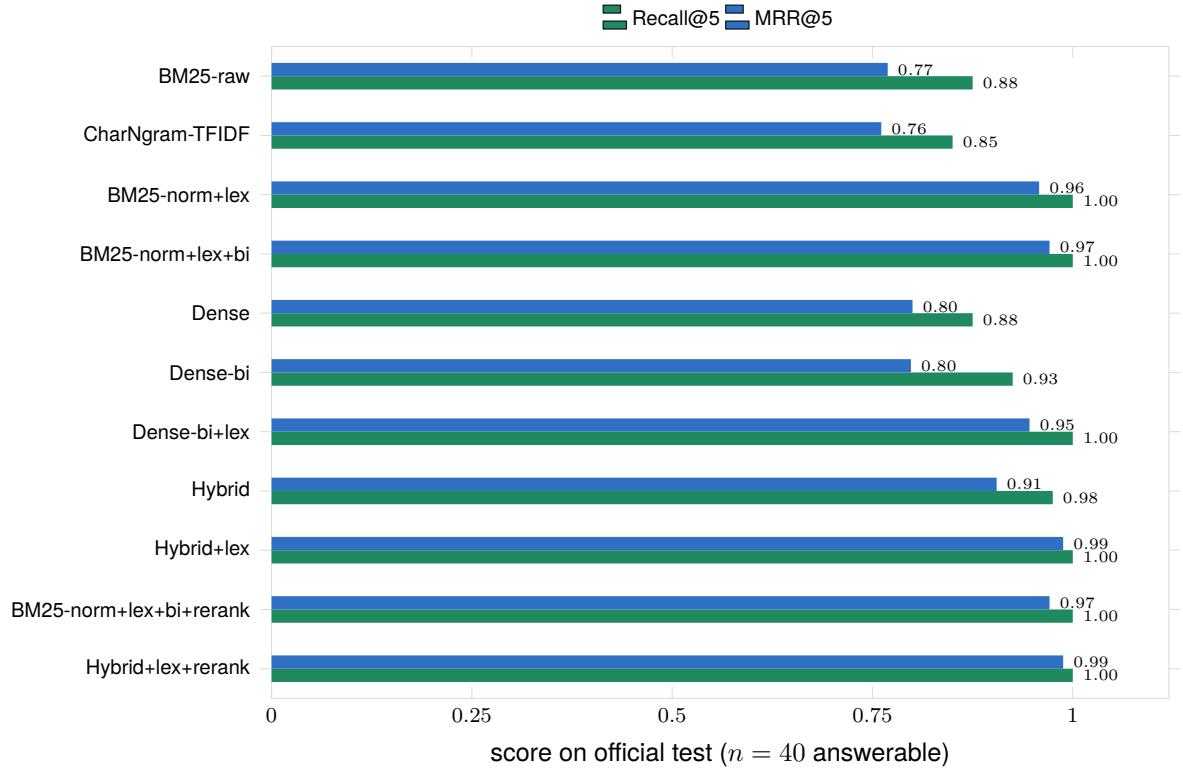


Figure 17. Retrieval on the official test split. The selected BM25-norm+lex reaches Recall@5 = 1.000 and MRR@5 = 0.958 against 0.875 / 0.769 for raw BM25. Dense e5 alone reaches Recall@5 = 0.875; combined with the lexicon (`Hybrid+lex`) it reaches the highest test MRR@5, 0.988, but that was not known when the choice was made on dev.

9.2 Cross-script consistency

For the 30 dev intents asked in all three scripts, the production retriever returns the same top passage in English, Urdu and Roman Urdu for 100.0% of intents, and all three hit the gold passage in their top 5 for 100.0%. For the 30 noisy Roman Urdu test variants, the top passage equals the English dev query's top passage 93.3% of the time. Raw BM25 and character n-grams score 0% on the same-top-1 test, because Urdu script shares no tokens with English text.

9.3 Blind set: the primary result

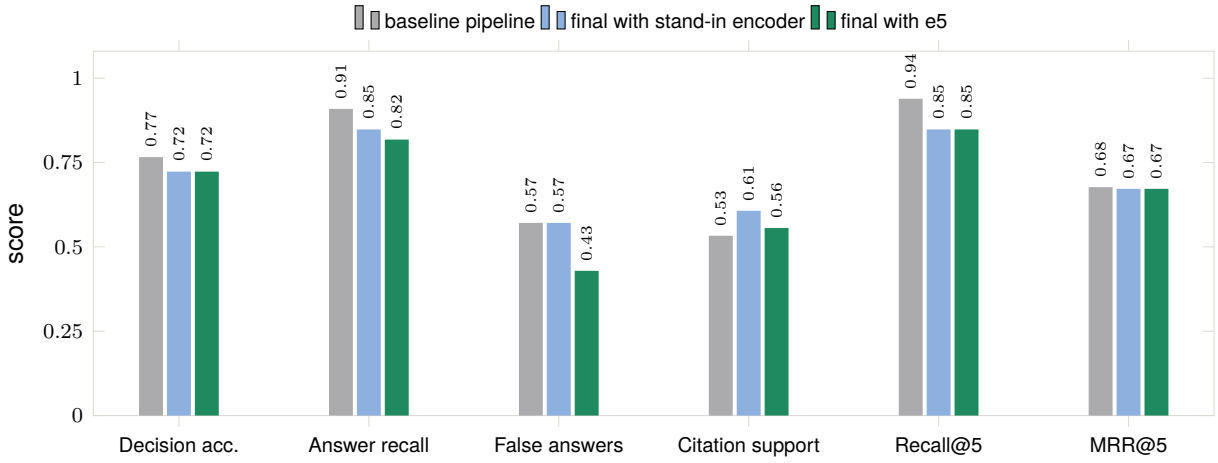


Figure 18. Blind set ($n = 47$; answerable $n = 33$, unanswerable $n = 14$). For "false answers" lower is better. With e5, false answers drop from 57.1% to 42.9% at the cost of answer recall (90.9% → 81.8%). All confidence intervals overlap at this sample size.

Blind metric	Baseline	Stand-in final	**e5 final**	n
Decision accuracy	76.6%	72.3%	**72.3%**	47
Answer recall	90.9%	84.8%	**81.8%**	33
False-answer rate (lower is better)	57.1%	57.1%	**42.9%**	14
Citation support	53.3%	60.7%	**55.6%**	27
Retrieval Recall@5	93.9%	84.8%	**84.8%**	33
Retrieval MRR@5	67.7%	67.2%	**67.2%**	33
Cross-language: all variants correct	83.3%	66.7%	**66.7%**	6 groups

Table 9. Blind results as numbers. The baseline is the pipeline before round-2 fixes.

9.4 Decisions across splits

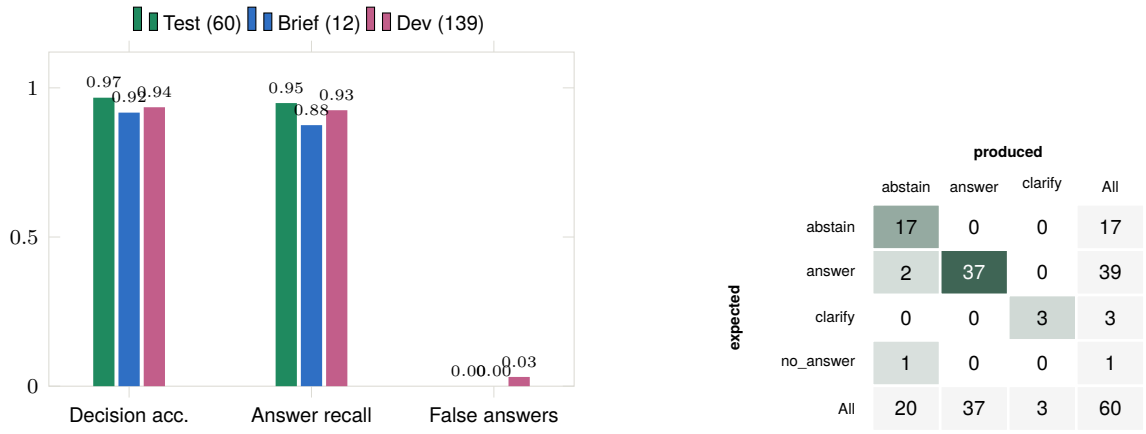


Figure 19. Left: decision accuracy, answer recall and false-answer rate per split. Right: expected versus produced decisions on the official test (darker = more queries). The one `no_answer` query (an unsupported generalization) was correctly not answered.

On the official test the e5 pipeline makes the right decision for 96.7% of queries, never answers an unanswerable one (0.0%), and its citations are supported 91.9% of the time. On dev its false-

answer rate is 3.1%, down from 9.4% with the stand-in encoder.

9.5 Latency

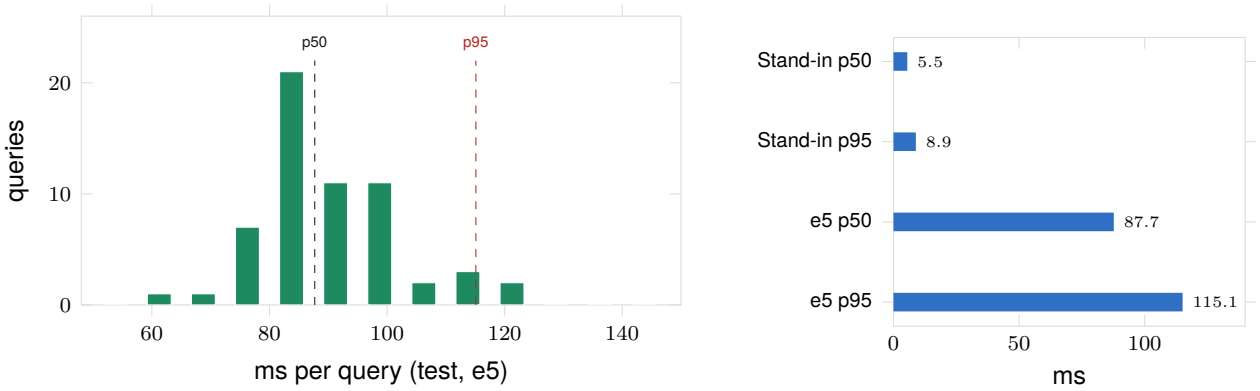


Figure 20. Response time on a laptop CPU (Intel Core i7-10510U). Embedding the query with e5 dominates: the median rises from 5.5 ms (stand-in) to 87.7 ms, still well below what a person notices. Repeated questions are cached and return in about 1 ms.

10 Error analysis

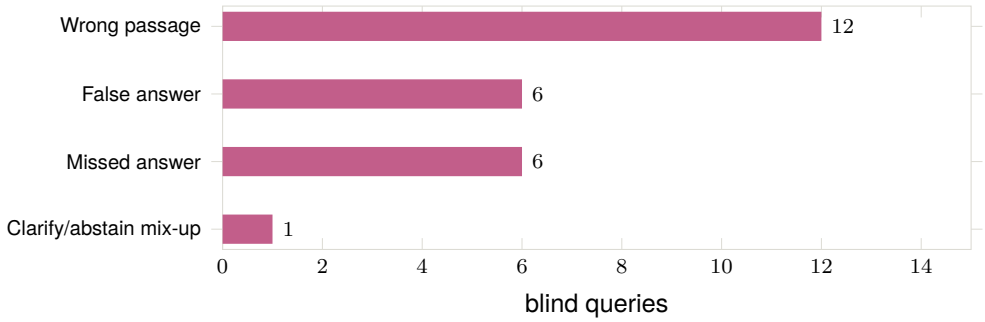


Figure 21. Error types on the blind set. "Wrong passage" means the decision was right (answer) but the cited passage was not the gold one; it is the largest category.

Official test and brief. The e5 pipeline makes 5 errors on the official test (3 confusable-sibling citations and 2 missed answers) and 2 on the brief's cases:

ID	Query	Gold	Got	Cause
N023	Kya 2 working days ka matlab passport ghar bhi 2 din mein aa jaye ga?	F15	F13	sibling: Normal vs. Fast Track timing
SV008	Minor k pasport k liye dono parents ki ID chahye?	F08	F07	sibling: two minor-document passages
SV018	Online renewal kis condition mein hoti hy?	F18	F17	sibling: online renewal vs. its eligibility
SV011	MRP 5 ya 10 saal ka hota hy?	F11	abstain	dense score below τ_d
SV023	Passport office mein interview hota hy?	F23	abstain	dense score below τ_d
BC04	Which documents are required to renew a passport?	F02/16/17/18	F26	"a passport" fuzzy-matched to "e passport"
BC07	pasport renu karwana hy	F16/17/18	abstain	dense 0.8169, just under $\tau_d = 0.8231$

Table 10. Every error on the official test and the brief. All are safe failures (a wrong-but-official quote or an abstention); none invents information.

Patterns.

- **Sibling passages** that differ by one word or number (F13/F14/F15, F07/F08, F17/F18) are the main retrieval failure. Concept re-ranking helps on dev but not enough on noisy Roman Urdu.
- **The e5 threshold is strict.** It cuts false answers but also abstains on some heavily misspelled questions, such as the brief's Case 5B. That is the trade the tuning objective chose (Figure 8).
- **A known bug.** The phrase-level fuzzy matcher reads "a passport" as "e passport" and adds the `EPASS` concept, which pulled the English renewal question (BC04) to F26. The fix (exact match on a phrase's first token) was found after the blind run, so applying it requires a fresh blind set.
- **Blind false answers** (6 of 14) come from topics the KB doesn't cover (contact details, travel rules, fees), an ambiguous question containing a topic word, and an injection phrased without the cue words.

10.1 Leakage audit

The lexicon and rule cues were written with the official test visible. Dropping the 19 lexicon forms that occur only in test queries lowers test Recall@5 from 1.000 to 0.975; dropping the 29 test-only cue phrases lowers test decision accuracy from 96.7% to 88.3% and raises the false-answer rate from 0.0% to 14.3%. These ablated figures are the conservative estimate for the official test, and they are why the blind set is treated as the primary result.

11 Stand-in encoder versus e5

	Stand-in (`841339e5e722`)	e5 (`d089d88a44a2`)
Dense threshold τ_d	0.6003	0.8231
Production retriever	BM25-norm+lex	BM25-norm+lex
Blind decision accuracy	72.3%	72.3%
Blind false-answer rate	57.1%	**42.9%**
Blind answer recall	84.8%	81.8%
Blind citation support	60.7%	55.6%
Dev false-answer rate	9.4%	**3.1%**
Official test decision accuracy	100.0%	96.7%
Median latency (test)	5.5 ms	87.7 ms

Table 11. What the real encoder changed.

With e5 the dense score separates answerable from unanswerable questions more cleanly, so the tuned threshold moves up and the system becomes more cautious: fewer false answers (the metric that protects citizens), slightly fewer answers overall. Ranking quality is unchanged because the production ranker is keyword-based; e5's main contribution is *knowing when not to answer*. The `Hybrid+lex` configuration, which fuses e5 with the lexicon, now scores highest on test MRR@5, which suggests it as the candidate to select once a fresh blind set exists.

12 Limitations and responsible deployment

- **Small corpus.** With 30 passages, retrieval resembles intent classification. Scaling to full DGI&P pages needs chunking and re-evaluation.
- **Hand-built rules.** The lexicon and gates were written by the same person who wrote the blind set; an independent test set is the next step.
- **No fee schedule.** Fees aren't in the knowledge base, so fee questions get a clarification or an abstention, never a number.
- **Freshness.** Official procedures change; every passage is marked "needs manual review" and must be re-checked against the live pages before any real deployment.
- **Translation.** The Urdu text is the organisers' translation; the English official excerpt stays authoritative.
- **Hardware.** Measured on a 16 GB laptop (Intel Core i7-10510U, CPU only). With the app, e5 and Qwen all loaded, the full evaluation run ran out of memory until Qwen was unloaded.
- **Not an official service.** The interface says so on every answer.

13 Practical impact

The assistant targets citizens who search in Roman Urdu or Urdu, including people with limited English, and gives them official, citable answers instead of forum hearsay. Refusing to guess about fees, guarantees or individual cases reduces the risk of misleading people about legal requirements. The architecture (normalize, bridge, retrieve, decide, verify) transfers to CNIC, licensing or other e-Sahulat services by swapping the corpus and extending the lexicon.

14 Future work

1. Write an independent blind set (by someone who hasn't seen the lexicon) and fix the "a passport" matcher bug against it.
2. Re-select the retriever with e5 on a larger dev set; `Hybrid+lex` is the leading candidate.
3. Add a passage-level check that the cited sentence states the specific claim asked about, to cut wrong-passage answers.
4. Expand the knowledge base from the live DGI&P pages through the notebook's review-and-approve ingestion step.
5. Self-host the web fonts so the interface renders fully offline.

15 Conclusion

E-Sahulat retrieves official passport evidence across Urdu script, Roman Urdu, English and code-mixed questions, answers only with verbatim, verified quotes, and says "not on record" when the evidence runs out. With the real e5 encoder it reaches 72.3% decision accuracy on a blind set while answering unanswerable questions 42.9% of the time, down from 57.1%. The numbers are modest at $n = 47$, and the report shows every failure, but each claim can be checked against the notebook run that produced it.

A Reproducing everything

Commands

```
pip install -r requirements.txt
python esahulat/run/run_notebook.py    (or run the notebook top to bottom)
python report/make_report_data.py
cd report && tectonic technical_report.tex
python server.py    then open http://127.0.0.1:8000
ollama pull qwen2.5:3b    optional, for the Qwen summary
```

File	Contents
`E_Sahulat_AI_Hackathon.ipynb`	the pipeline, tuning and evaluation (cells tagged `pipeline` are what the app runs)
`server.py`, `web/`	the web application
`llm.py`	Ollama client, guardrails, number check, start-up helper
`esahulat/outputs/`	per-query results, error analysis, blind results, `app_metrics.json` (generated)
`FINAL_EVALUATION_REPORT.md`	the run's own summary of every metric
`report/make_report_data.py`	turns the outputs into this report's numbers and charts
`DESIGN.md`, `PRODUCT.md`	the interface's design system and product notes

Table 12. Where everything lives.

Dataset fingerprint: SHA-256 prefix `5e0e45bce102a95a`; blind set: `2d821119487a`; random seed 42.